



SWATI GOYAL

PhD Candidate
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DOB: 30/04/1997
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PROFESSIONAL DESCRIPTION

PhD candidate in Materials Science with research experience in semiconductor thin-film growth, process development, and device engineering. My work focuses on CVD/PVD-based thin-film deposition, growth kinetics and reaction mechanisms, vacuum processing, flexible electronics, and semiconductor device development, integrating experimental analysis with process modelling.

EDUCATION

- PhD in Materials Science** July 2022- Present
Indian Institute of Technology Bombay, Mumbai CPI: 9.83/10
Thesis title: Development of Amorphous-Silicon-based Flexible Alpha-Particle Detectors by Pulsed HWCVD
PhD Supervisors: Prof. Triratna P. Muneshwar and Prof. Rajiv O. Dusane
- Master of Science (M.Sc. Physics)** 2018-2020
Department of Physics & Astrophysics, Ramjas College, University of Delhi, Delhi 68.5%
- Bachelor of Science (B.Sc. Physics)** 2015-2018
Atma Ram Sanatan Dharma College, University of Delhi, Delhi CPI: 8.72/10

RESEARCH EXPERIENCES

- Thin-Film Device Engineering:** Fabrication and development of flexible a-Si:H p-i-n devices for alpha-particle detection, with emphasis on device architecture, leakage-current suppression, and process reproducibility.
- Pulsed HWCVD Process Development:** Developed a Pulsed HWCVD approach for a-Si:H film, demonstrating a stable growth rate, higher precursor utilization, suppressed filament silicide formation, and delamination-free film growth.
- Growth & Reaction Kinetics:** MATLAB-based modelling of a-Si:H growth and filament reaction kinetics, incorporating surface reaction pathways, species evolution, and silicide formation mechanisms.
- Ta-Si Silicide & Filament Ageing:** Investigation of Ta-Si silicide formation and filament ageing during HWCVD, with emphasis on understanding mechanisms and developing a pulsed deposition approach to suppress silicide growth and filament ageing.
- Interface Engineering:** Deposition and investigation of hydrogenated silicon carbide (a-SiC:H) as an interfacial and blocking layer in a-Si:H p-i-n devices.
- Thin-Film Characterization:** Electrical, optical, structural, and compositional analysis using I-V, CPM, Hall measurement, UV-Vis, FTIR, Raman, SEM, EDX, WDS, XPS, profilometry, and MOS.
- Thin-Film Stress Engineering:** Investigation of film stress evolution, delamination, and stress-relaxation mechanisms in thick a-Si:H films.
- SHJ & p-i-n Solar Cell Fabrication:** Fabrication and Characterization of Silicon heterojunction (SHJ) and p-i-n solar cells, including ITO transparent conducting oxide deposition by RF magnetron sputtering.
- Semiconductor Processing Platform Integration:** Integrated a multi-chamber semiconductor processing platform comprising HWCVD, RF magnetron sputtering, and load-lock modules. Established vacuum, process-gas delivery, cooling, exhaust, and utility interfaces, and developed the associated process conditions for thin-film growth.

TECHNICAL SKILLS

- Process & Fabrication Equipment:** Hot-Wire Chemical Vapor Deposition (HWCVD) Cluster Tool, Plasma-Enhanced Chemical Vapor Deposition (PECVD), DC and RF Sputtering, Thermal Evaporation, Multi-pocket Electron-Beam Evaporator, Reactive Ion Etching (RIE), Spin Coating, Manual Mask Aligner MJB4.
- Process Safety & Vacuum Systems:** Handling of pyrophoric and toxic process gases; installation, maintenance, and troubleshooting of vacuum systems, including rotary pumps, pressure gauges, and mass flow controllers (MFCs); RGA-based contamination analysis and helium leak testing.
- Device and Material Characterization:** Keithley Model 6517B Electrometer, 4200-SCS Parameter Analyzer, Polaris (Stress/ long time measurements), Four-Probe I-V Measurement, AAA Solar Simulator, CPM, Hall Measurement, Profilometer, UV-Vis Spectroscopy, SEM, EDS, FTIR, Raman Spectroscopy, GDOES, QSSPC.
- Software & Data Analysis:** MATLAB, Origin, CorelDRAW, Inkscape, Gwyddion, ImageJ, and CasaXPS

PUBLICATIONS

- **Swati Goyal**, Karam Veer Singh, Rajiv O. Dusane and Triratna P. Muneshwar, “Pulsed Hot-Wire CVD of a-Si:H films: from catalytic SiH₄ dissociation model to growth experiments - *Under Review*.
- Karam Veer Singh, **Swati Goyal**, Sapna Chandaul and Rajiv O. Dusane, “Microstructure-Dependent Response of HWCVD a-Si:H Passivation Layers to Subsequent a-SiN:H Deposition: Implications for Integrated Optical and Surface Passivation - *Under Review*.
- Rajib Kar, Akash Barnwal, N.N. Patel, **Swati Goyal**, Vanita Sekar, D.S. Patil, R.O. Dusane, and R.L. Bhardwaj, “Microwave Cold Plasma Synthesis of Carbon Nanostructures on Stainless Steel Substrates with Changing Plasma Chemistry” in SEBTA-2024.
- Rohini Sanikop, Diwakar Singh, Ramesh Rajarapu, **Swati Goyal**, Saurabh Lodha, Deepo Kumar, Triratna Muneshwar, and Tanushree Choudhary, “Influence of growth parameters in powder source chemical vapor deposition of MoO₂”-*Submitted*.

INVITED TALKS AND WORKSHOPS

- Speaker at the “HoFCVD Thin-Film Equipment and Application Technology Conference & 12th International Conference on Hot-Wire CVD”, 2026, China.
- Speaker at the “International Symposium on Semiconductor Materials and Devices (ISSMD-2026)”, IIT Ropar, India.
- Poster Presentation: Presented PhD research work at the MEMS-ASPIRE Industry-Academia Meet 2026, IIT Bombay.
- Participation in a workshop on “Radiation detectors: Technology Development towards Flexible Devices, 2024, IIT Bombay, India.

INTERNATIONAL COLLABORATION & RESEARCH VISITS

School of Physics, Engineering and Technology, University of York, United Kingdom-collaborative research with Dr. Samadhan Patil and Prof. David Jenkins.

- **Research Visit-June 2025**: Trained in the safe handling of alpha radioactive sources; performed alpha-particle characterization of standard crystalline-Si detectors.
- **Research Visit-February 2026**: Performed alpha-particle response characterization of a-Si:H p-i-n test structures using a ²⁴¹Am alpha source; conducted photolithographic processing and device patterning in a cleanroom using a KLOE mask-aligner system.
- **Research Training & Knowledge Transfer-2024**: Trained a visiting M.Tech student from the University of York in CVD/PVD thin-film deposition and material characterization.

OTHER COLLABORATIONS

- **Prof. Dipti Gupta, IIT Bombay**: Collaborative electrical characterization and I-V measurements of a-Si:H p-i-n device structures.
- **Prof. Rudrajyoti Palit, TIFR Mumbai**: Collaborative alpha-particle characterization of a-Si:H test structures using different alpha sources.
- **Prof. Dinkar Patil, BARC Mumbai**: Collaborative alpha-particle characterization of a-Si:H test structures.

PROJECTS & SEMINARS

- **PhD Seminar**: Plasma-Enhanced Atomic Layer Deposition (ALD) *July 2020-Dec 2020*
Supervisor: Prof. Dinkar Patil
Studied ALD process schemes and precursor selection, with emphasis on thermal ALD and plasma-enhanced ALD (PEALD)
- **M.Sc. Dissertation**- Multicomponent Oxide Thin Films for Energy Harvesting *Jan 2020-Oct 2020*
Development of miniaturized MEMS cantilever structures for mechanical-vibration-based energy harvesting.
Supervisors: Prof. Vinay Gupta, Prof. Monika Tomar, Dr. Reema Gupta

KEY COURSES

Plasma Processing of Materials || Physics of Materials || Growth and Characterization of Nano-electronic Materials || Materials & Processes for Semiconductor Devices || Semiconductor Photochemistry & Photocatalysis || Science and Technology of Thin Films || Introduction to Ab-Initio Methods in Materials Modelling

SCHOLASTIC ACHIEVEMENTS

- IIT JAM (Physics), 2018: Secured All-India Rank of 907
- JEST (Physics), 2022: Secured All-India Rank of 301
- GATE (Physics), 2022: Secured All-India Rank of 1576
- B.Sc. Physics, 2017: Secured 2nd position in College | CGPA: 9.25/10
- Matriculation, 2013: Secured 1st position in School | CGPA: 9.6/10

POSITIONS OF RESPONSIBILITIES

- Teaching Assistant, Semiconductor Thin Films Laboratory | 2022-2023
- Teaching Assistant, Glow Discharge Optical Emission Spectroscopy (GDOES-HORIBA) facility | 2023-2025
- Teaching Assistant, Scanning Electron Microscopy (SEM-JEOL) facility | 2024-2026
- Teaching Assistant, Electronic, Magnetic and Optical Properties of Materials | 2025
- Teaching Assistant, Science and Technology of Thin Films | 2026-Present
- Represented the Semiconductor Thin Films & Plasma Processing Laboratory at IIT Bombay Techfest | 2022-2025
- Workshop Coordination, IIT Bombay & University of York; Assisted in organizing two workshops on Radiation Detectors | August 2024 & January 2025
- Vice President, Department of Physics, Atma Ram Sanatan Dharma College, University of Delhi; conducted FEYMANIA 2020, a departmental Physics outreach and academic event | 2017-2018.

REFERENCES

Prof. Triratna Muneshwar

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Prof. Rajiv O. Dusane

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